IQRA FAREED

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Software Engineer with 5+ years of experience building scalable web applications and cloud-connected systems with expertise in modular component design, frontend architecture, API integrations, and performance optimization. Currently pursuing an MSCS with a research focus on causal representation learning for domain-generalizable medical image classification and uncertainty quantification in LLMs.

Research Experience

University of Louisiana at Lafayette | Graduate Researcher, Medical AI Lab (Dr. Min Shi) | *April 2026 - Present*

- Designed and implemented a sensitivity_at_specificity() evaluation metric for clinically aligned model assessment (sensitivity at fixed 90% specificity)
- Developed **DCGM**, a novel causal domain-generalization method extending CaSN, adapting components from a related fairness-focused framework (MultiFair)
- Built and managed GPU training pipelines (PyTorch, ResNet-50, NVIDIA A6000) across two distinct imaging domains (hospital + mobile-captured retinal fundus images) for domain-generalizable diabetic retinopathy detection

Uncertainty Quantification in LLMs Survey Research *July 2026 - Present*

- Authoring a taxonomic comparative survey on Uncertainty Quantification in LLMs, across Detection, Mitigation, and Human-Alignment categories
- Built a structured literature taxonomy and comparative matrix to organize and evaluate methods across token-level, semantic-entropy, and calibration-based UQ approaches

Professional Experience

TripShepherd

Senior Software Developer *May 2024 - August 2025*

- **Migrated the master app from React to Next**, achieving a **70% improvement** in page performance.
- Designed a modular frontend architecture integrated with REST APIs and real-time data services.
- Implemented structured state management and asynchronous data handling to ensure consistency across complex workflows.
- Maintained unit tests to validate component behavior and ensure reliable feature releases.

Qisstpay

Software Developer *September 2022 - August 2023*

- Developed scalable SPAs and feature modules using React.js and Next.js, defining Redux-based state management patterns.
- Implemented responsive UI systems using Material UI and Tailwind CSS, ensuring accessibility and consistency.
- Integrated complex **REST APIs** built in **Java and Go**, collaborating closely with backend teams on data contracts.

Naxxa Consulting

Associate Software Engineer, *November 2020 – August 2022*

- Developed responsive healthcare and enterprise web applications using React.js and Material UI, improving usability and consistency across patient-facing and internal workflows.
- Implemented real-time features using **AWS AppSync GraphQL subscriptions** and REST APIs via **Axios**.
- Developed scalable telehealth and enterprise UI workflows focused on accessibility and real-time user interactions.

Additional Experience

Graduate Teaching Assistant, Dept. of Engineering Management, UL Lafayette (Aug 2025–Aug 2026) ·
Graduate Research Assistant, Center for Louisiana Studies, UL Lafayette (Aug 2026–Present)

Skills

React.js, Next.js, JavaScript, Tailwind, TypeScript, Python, Node.js, Express.js, Machine Learning, Medical Imaging, PyTorch
Web Optimization Techniques, WebSockets, AWS, Firebase, Tailwind CSS, REST API, GraphQL, TanStack Query (React Query), Causal Inference

Notable Projects

- TelyMd Project: Built a healthcare consultation platform using React.js and Material UI, focused on responsive patient-facing workflows, appointment experiences, and intuitive healthcare UI systems.
- Boat App: A dynamic web portal developed with Next.js, featuring real-time bookings, vehicles, and other functionalities.

Education

University of Louisiana at Lafayette, USA
Master of Science in Computer Science (MSCS); ***CGPA:4.0*** *August, 2025 - December, 2026*
COMSATS University Islamabad, Pakistan
Bachelor of Science in Software Engineering (BSSE) *February 2017 - December 2021*